

5(a)	any two from: 1 most water (in xylem) goes to the leaves taking (dissolved) phosphate ions with it ; 2 xylem tissue does not extend into, growing points / shoot tips / root tips ; 3 higher demand for phosphate ions in leaves (for photosynthesis) ; 4 water and phosphate ions are transported upwards in the xylem (not down to root tips) ; 5 phosphate ions are absorbed in area above root tips (so don't go down) ; A transport to root tips involves movement in phloem 6, 7 AVP ;; e.g. correct ref to blockage by, endodermis / Casparian strip / suberin (apoplast pathway) not all phosphate ions cross cell (surface) membrane of endodermal cells (symplast pathway) phosphates may be stored in (vacuoles of cells in), roots / stems phosphate ions are used in the root for making, ATP / nucleotides / AW	2
5(b)	any four from: 1 phosphate ions transported in the phloem move, out of the leaf / into stem ; 2 (group B data suggests that) phloem transports (some of) phosphate ions downwards ; 3 phosphate ions can move from phloem to xylem ; 4 waxed paper prevents movement (between phloem and xylem) ; ora 5 data in support of difference between waxed and unwaxed in the same sample area ; e.g. in S1 1% in xylem in A as opposed to 5% in xylem in B / <u>percentage</u> in xylem always higher in group B 6 some phosphate ions may transfer to surrounding cells in the stem ; 7 much of the phosphate may not have left the leaf / much of the phosphate has been transported away from the leaf in the hour ; 8 AVP ; e.g. some may travel upwards in the phloem but no data not clear that results show phosphate ions transported upwards in xylem <i>ref. to</i> anomalous results / results not conclusive not all phosphate ions injected are accounted for	4